Task Description:

You are tasked with developing an algorithm to automatically identify seismic "first breaks" from provided seismic trace data. First breaks are the initial arrivals of seismic waves, which are critical for processing and interpreting seismic data, especially in seismic refraction and tomography studies. The provided dataset includes labeled seismic traces, where each trace is annotated with the time corresponding to its first break. Your goal is to create a robust, automated method that can accurately detect these first breaks across different seismic assets, regardless of varying signal-to-noise ratios and variation of first breaks line behavior across the different seismic assets.

Data Overview:

The provided dataset consists of seismic traces from four different real-world seismic assets, stored in HDF5 format. Each trace in the dataset is manually labeled with first break times where available. Your objective is to develop an automated method that accurately identifies these first breaks.

The dataset contains the following four seismic assets, available for download via the following links:

- Brunswick: https://d3sakqnghgsk6x.cloudfront.net/Brunswick_3D/Brunswick_orig_1500ms_V2.hdf5.xz
- Halfmile: https://d3sakqnghgsk6x.cloudfront.net/Halfmile_3D/Halfmile3D_add_geom_sorted.hdf5.xz
- Lalor: https://d3sakqnghgsk6x.cloudfront.net/Lalor_3D/Lalor_raw_z_1500ms_norp_geom_v3.hdf5.xz
- Sudbury: https://d3sakqnghgsk6x.cloudfront.net/Sudbury_3D/preprocessed_Sudbury3D.hdf.xz

Within each HDF5 file, the seismic data is stored in the “/TRACE_DATA/DEFAULT” group, and the data can be explored using:

Python code:

```
import h5py

print(dict(h5py.File(path_to_file)['TRACE_DATA/DEFAULT']))
```

The following keys could be useful for the task:

- data_array - 2D array of all recorded seismic traces (trace count x sample count)
- SHOTID - shot identifiers (one entry per trace)
- SHOT_PEG - shot PEG identifiers (one entry per trace)
- SOURCE_X - X coordinate of the shot location (one entry per trace)
- SOURCE_Y - Y coordinate of the shot location (one entry per trace)
- SOURCE_HT - height/elevation of the shot location (one entry per trace)
- REC_PEG - PEG identifiers for the receivers (one entry per trace)
- REC_X - X coordinate of the receivers (one entry per trace)
- REC_Y - Y coordinate of the receivers (one entry per trace)
- REC_HT - height/elevation of the receivers (one entry per trace)
- SAMP_RATE - sampling rates in microseconds (one entry per trace, but should be constant)
- COORD_SCALE - scale factors to be applied to coordinates (one entry per trace, but should be constant)
- HT_SCALE - scale factors to be applied to heights/elevations (one entry per trace, but should be constant)
- SAMP_NUM - sample counts (one entry per trace, but should be constant)
- SPARE1 - first break time values (in msec). If the value is 0 or -1, the corresponding trace is considered as unlabeled.

Task Instructions:

The provided data consist of sequence of 2D seismic images (Figure 1). You should first reorganize the dataset into a format where each 2D seismic image (a sequence of traces) is separated. You can use the coordinate of the receivers to do this. Each seismic trace in the image is accompanied by its manually labeled first break pick (Figure 2). After splitting, you should have pairs of 2D seismic images and their corresponding first break labels (e.g., Figure 3).

You are free in choose any approach to develop the solution for this task. You should prepare the short presentation (around 25-35 minutes) that includes brief data overview, a detailed description of your methodology, its results and validation. At the end, you should submit the final code and the prepared presentation.

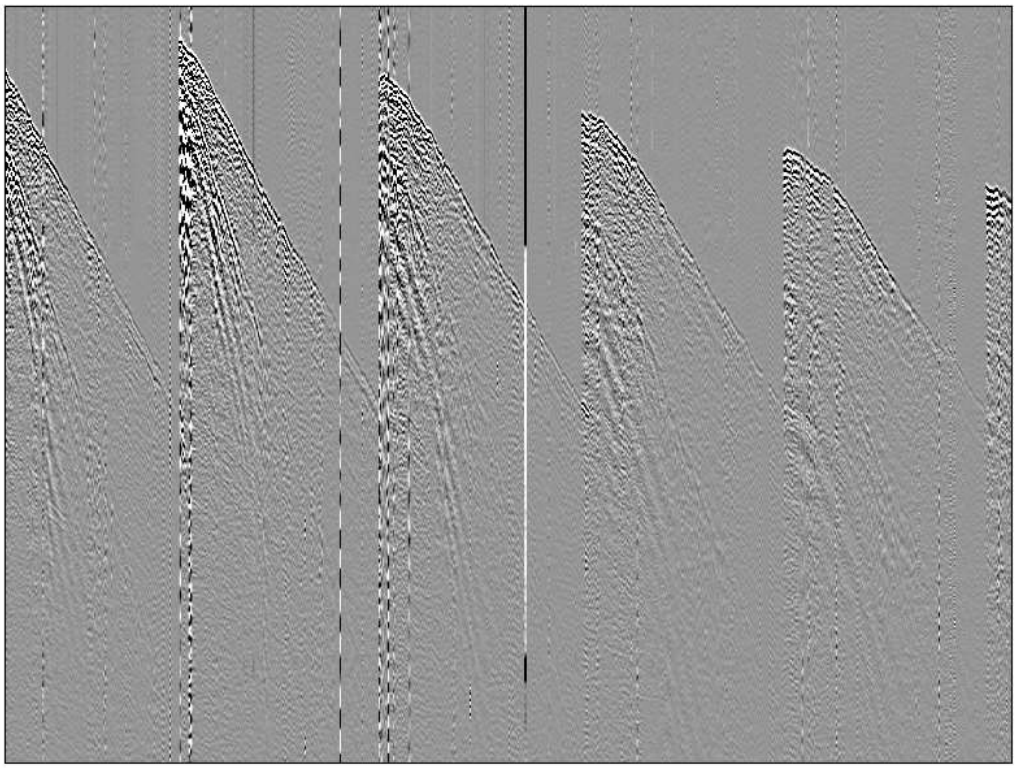

Figure 1. Example of 2D seismic images sequence from one of the provided datasets. X-axis is the seismic trace, Y-axis is the time.

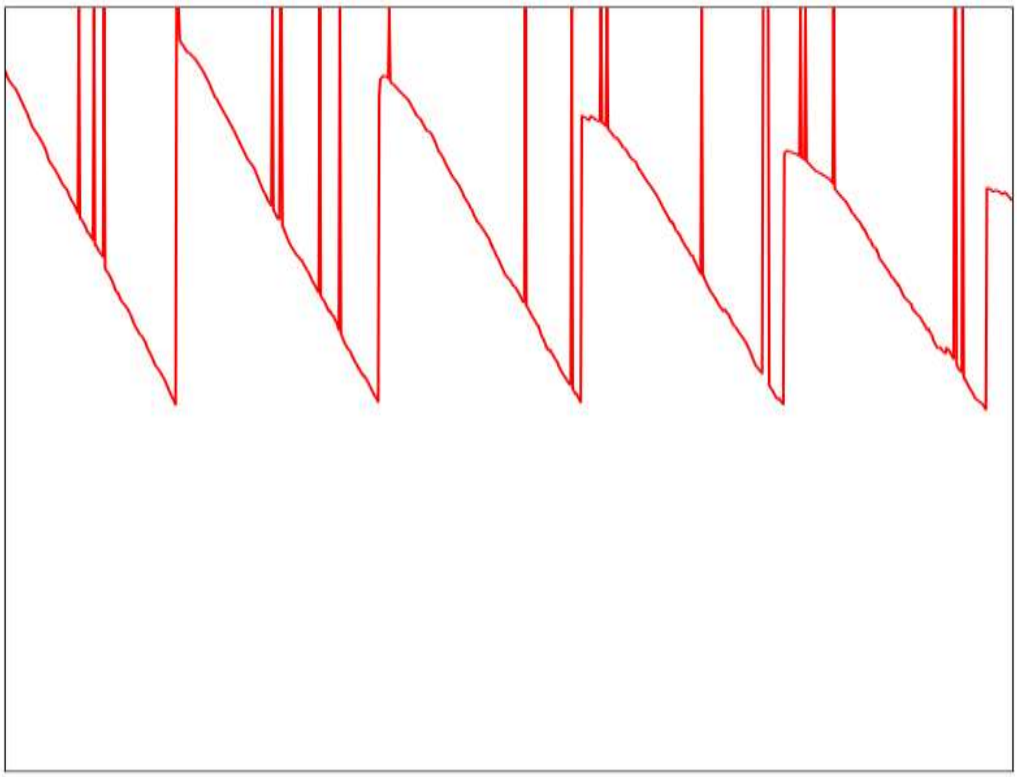

Figure 2. Corresponding first break picks for the Figure 1.

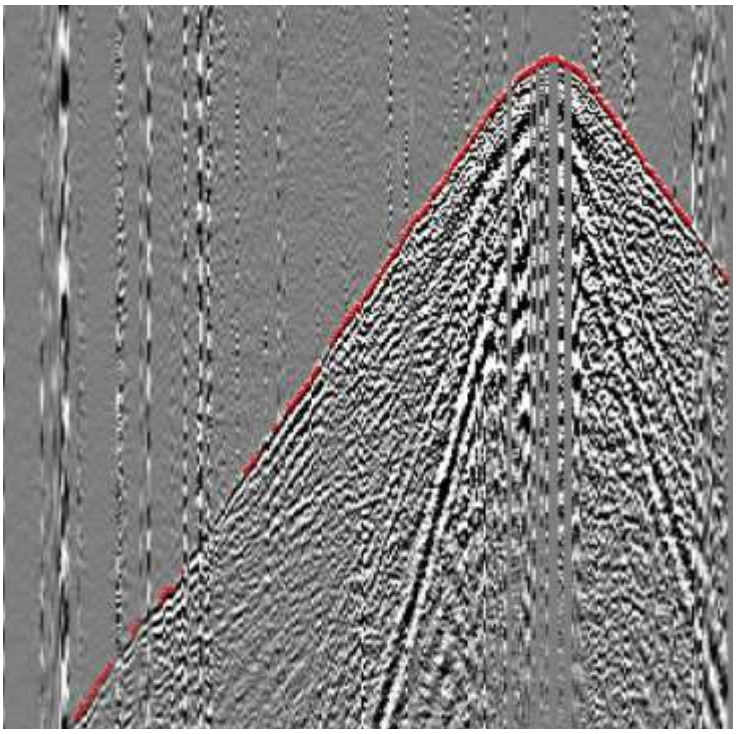

Figure 3. Example of 2D seismic image with first break line in red.